Table of Contents

1. Assignment GitHub Repo

Git hub URL: https://github.com/awsbiplab/pythonassignment.git

All Charts folder: This has all the plotted charts and graphs for the assignment

https://github.com/awsbiplab/pythonassignment/tree/main/reports/figures

2. Deliverable

Below are some to the key insights from the data set provided

2.1 Key Insights Report

2.1.1 Data Quality and Preparation

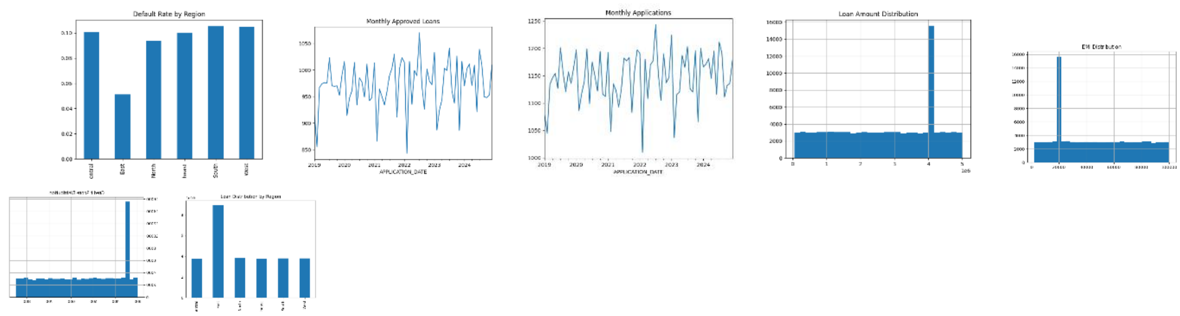
- Check for missing values, duplicate entries, and inconsistent data
- Standardize date formats and remove irrelevant columns.
- Handle outliers in numeric columns like Loan_Amount, Interest_Rate, and Default_Amount.

2.1.2 Descriptive/Exploratory Data Analysis.

2.1.2.1 Insight:

- Uniform credit core distribution.
- Emi and loan show clustering.
- Applications and approvals including rejections are stables.
- Default varies by region.

2.1.2.2 Visualization:



Note: charts reference:

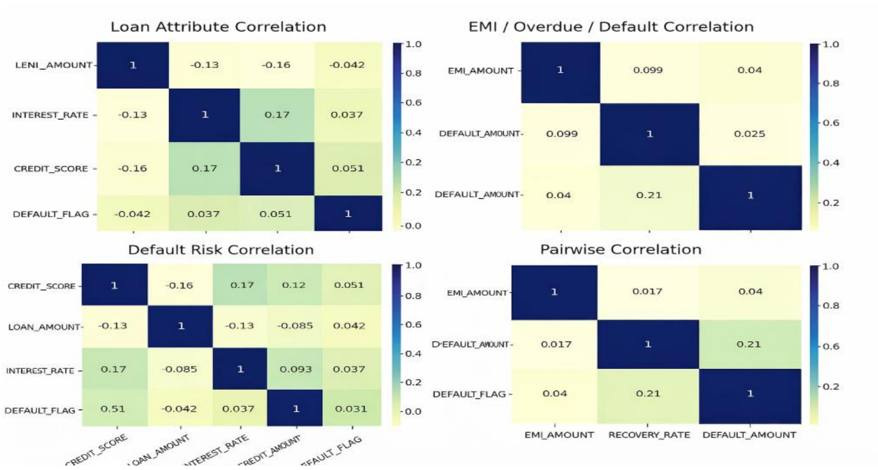
- task_2_credit_score.png
- task_2_emi_distribution.png
- task_2_loan_distribution.png
- task_2_monthly_applications.png
- task_2_monthly_approved.png
- task_2_region_default.png

2.1.3 Correlation Analysis

2.1.3.1 Insight:

- Correlation between loan attributes and defaults.
- EMI, overdue, and default amounts show some relationship.

2.1.3.2 Visualization:



Note: charts reference:

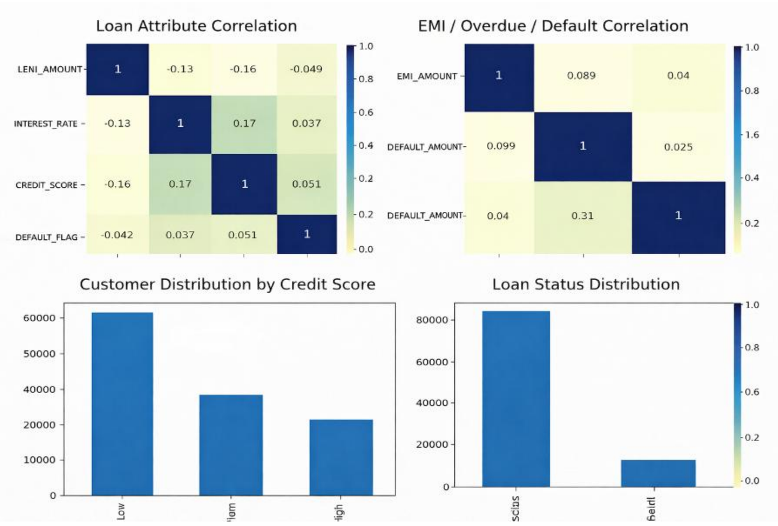
- task_3_correlation_main.png
- task_3_correlation_pairwise.png
- task_6_adv_default_corr.png
- task_6_adv_pairwise_corr.png

2.1.4 Customer Segmentation.

2.1.4.1 Insight:

- Hight Risk Customer are “Low income and low credit score”.
- Hight Value Customer are “High income and High credit score”.
- Customer Segmentation helps in identifying High-value and High-risk Customers
- Applications and approvals including rejections are stables.

2.1.4.2 Visualization:



Note: Charts reference:

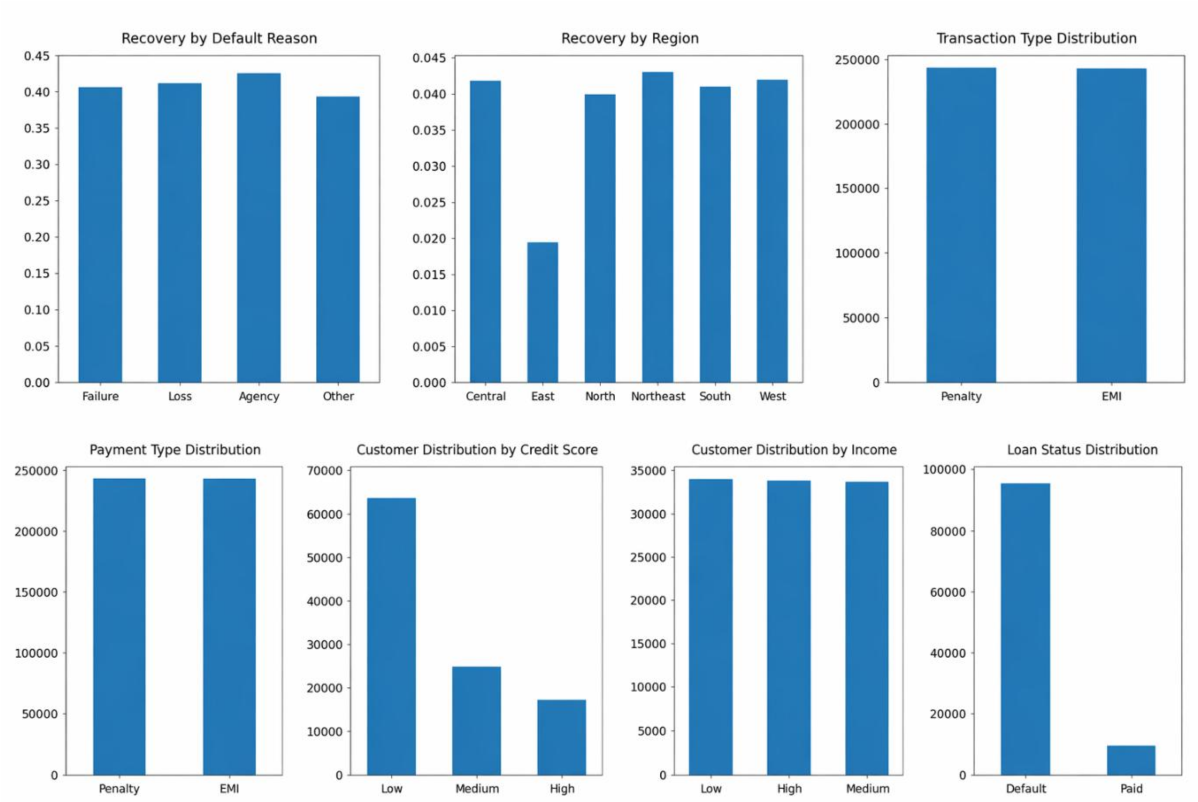
- task_5_income_segment.png
- task_5_credit_segment.png
- task_5_loan_status.png

2.1.5 Transaction & Recovery Analysis

2.1.5.1 Insight:

- Transaction patterns show customer repayment behaviour.
- Recovery rates vary by region.

2.1.5.2 Visualization:



Note: charts reference:

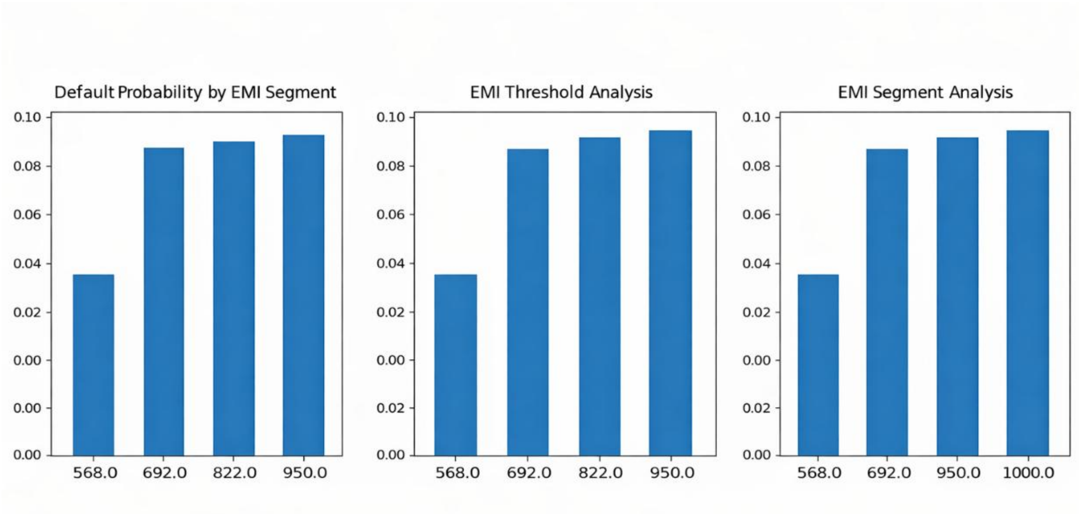
- task_7_txn_type.png
- task_7_recovery_reason.png
- task_7_recovery_region.png
- task_20_txn_type_dist.png

2.1.6 EMI Analysis

2.1.6.1 Insight:

- Higher the EMI higher is the probability of defaulting.

2.1.6.2 Visualization:



Note: charts reference:

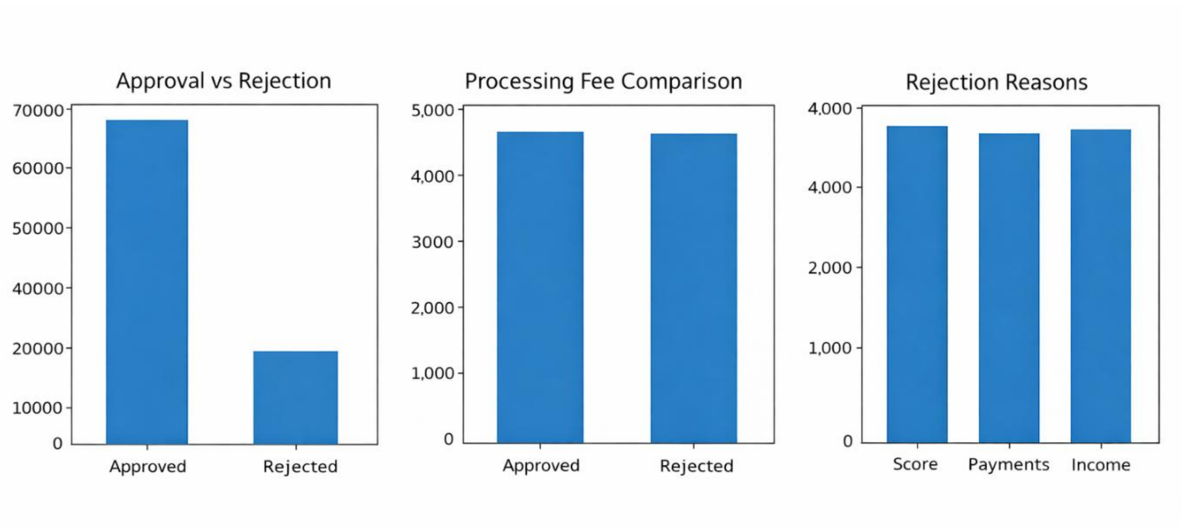
- task_8_emi_default.png
- task_8_emi_threshold.png

2.1.7 Lone Application Insights

2.1.7.1 Insight:

- Approval rate is high but selective
- Rejections driven by risk factors
- Fees vary based on approval status.

2.1.7.2 Visualization:



Note: charts reference:

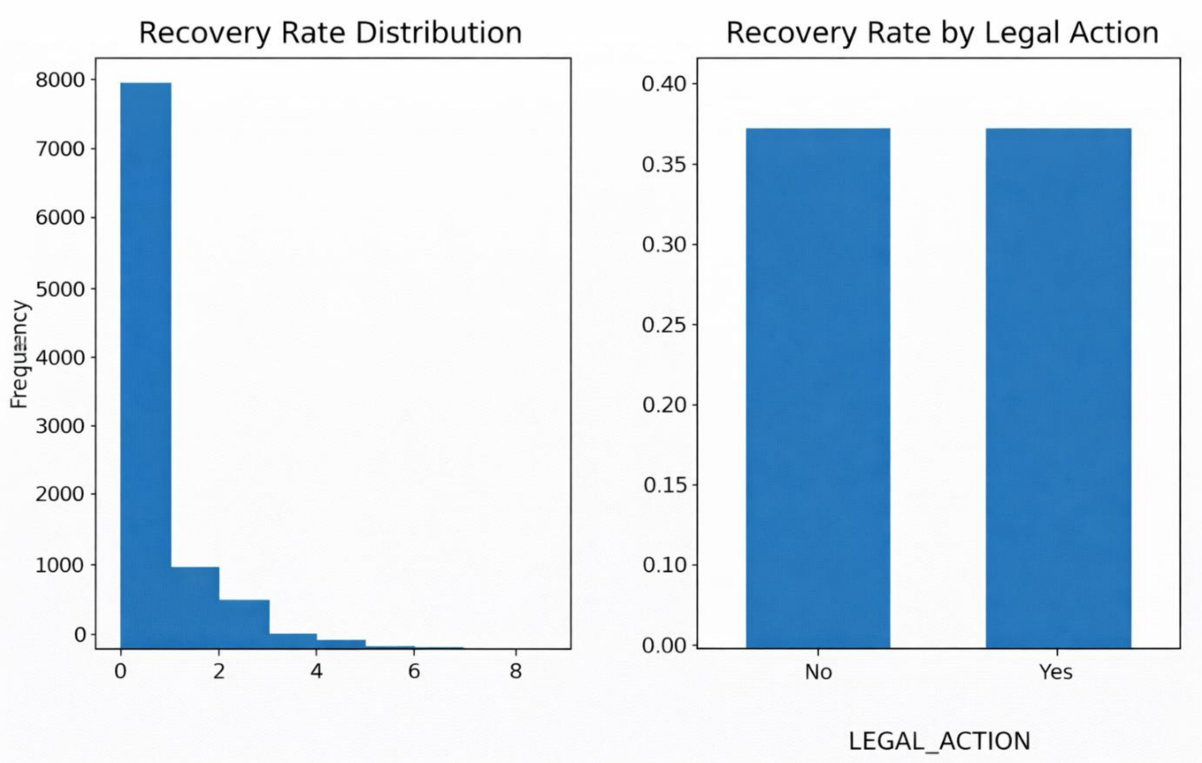
- task_9_application_status.png
- task_9_rejection_reason.png
- task_9_processing_fee.png

2.1.8 Recovery Effectiveness

2.1.8.1 Insight:

- Recovery rate improves with legal action
- Some loan types take longer mean bottlenecks.

2.1.8.2 Visualization:



Note: charts reference:

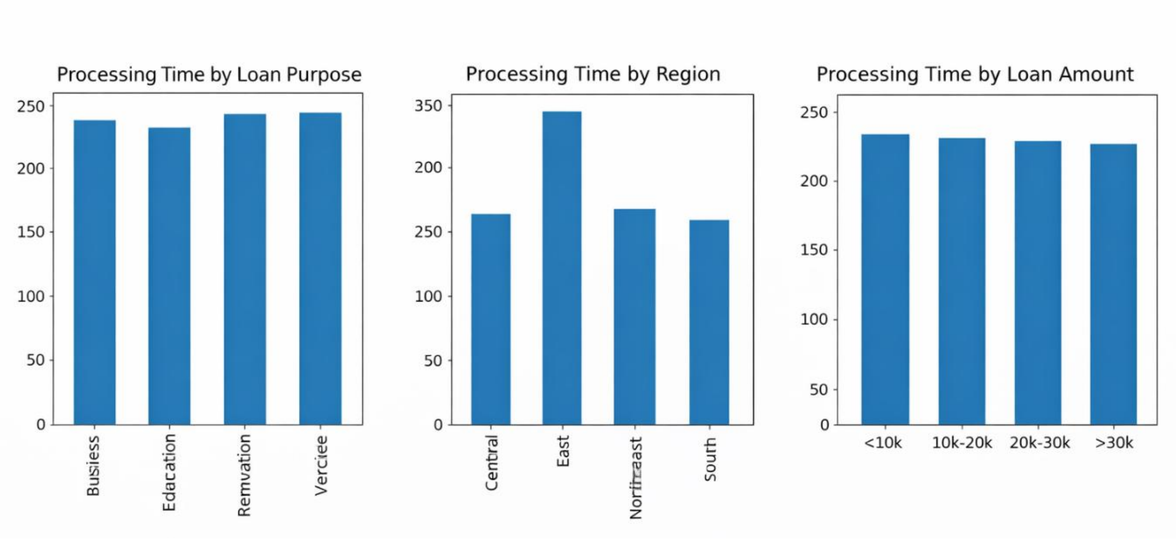
- task_10_recovery_legal.png
- task_10_recovery_rate.png

2.1.9 Loan Disbursement Efficiency

2.1.9.1 Insight:

- Processing time varies across regions
- Some loan types take longer → bottlenecks
- Data cleaning required for accurate analysis

2.1.9.2 Visualization:



Note: charts reference:

- task_11_disbursement_region.png
- task_11_disbursement_purpose.png

2.1.10 Profitability Analysis

2.1.10.1 Insight:

- Processing time varies across regions
- Some loan types take longer → bottlenecks
- Data cleaning required for accurate analysis

2.1.10.2 Visualization:

Note: charts reference:

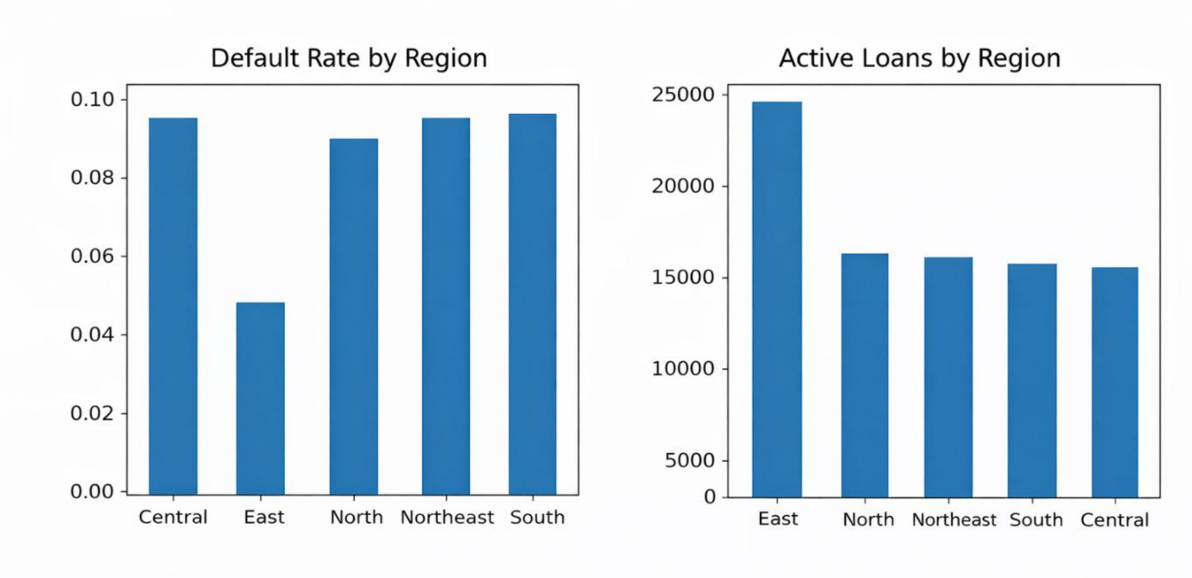
- task_12_profit_purpose.png
- task_12_profit_region.png

2.1.11 Geospatial Analysis

2.1.11.1 Insight:

- Profit varies by loan purpose
- Some regions contribute more revenue

2.1.11.2 Visualization:



Note: charts reference:

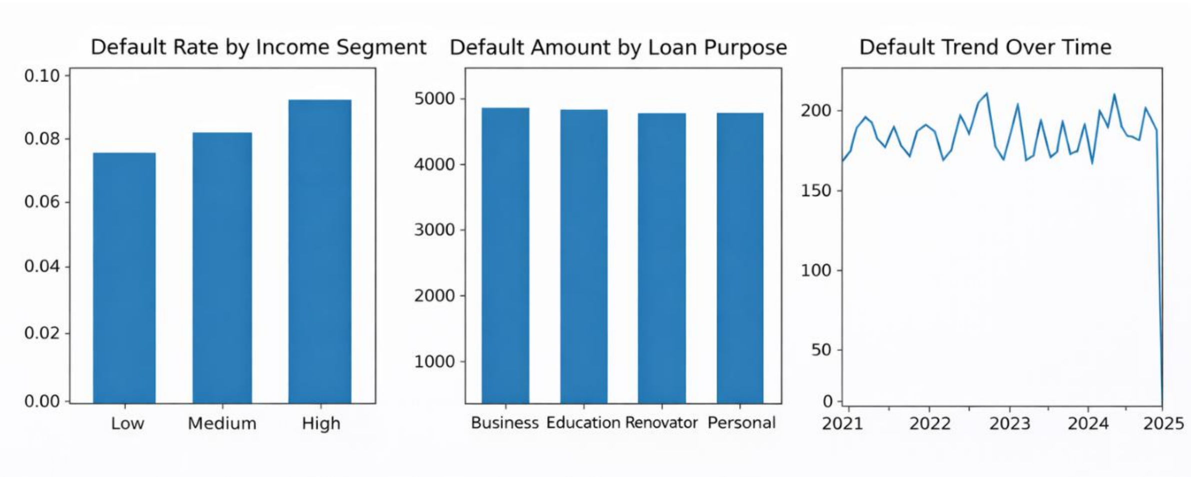
- task_13_geo_distribution.png
- task_13_geo_default.png

2.1.12 Default Trends Analysis

2.1.12.1 Insight:

- Certain loan purposes have higher default
- Low-income segments are more vulnerable

2.1.12.2 Visualization:



Note: charts reference:

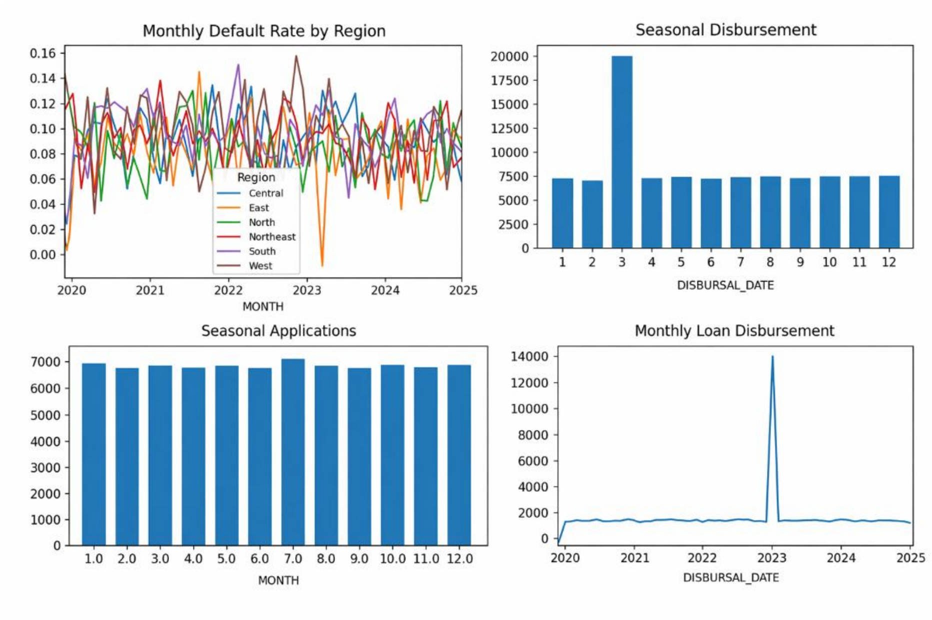
- task_14_default_trend.png
- task_14_default_purpose.png
- task_14_default_income.png

2.1.13 Time-Series Analysis

2.1.13.1 Insight:

- Loan demand is steady with time.
- Default rates fluctuate over time.
- Regional trends visible.

1.1.13.2 Visualization:



Note: charts reference:

- task_16_time_disbursement.png
- task_16_time_seasonal_app.png
- task_16_time_seasonal_disb.png
- task_16_time_default_region.png

2.1.14 Customer Behaviour Analysis

2.1.14.1 Insight:

- Repayment behaviour identifies risk early.

1.1.14.2 Visualization:

Note: charts reference:

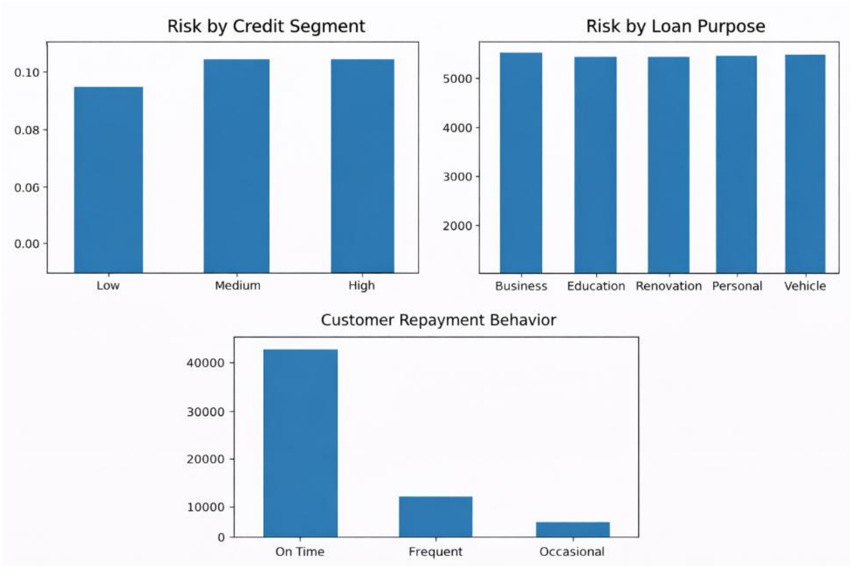
- task_17_customer_behavior.png
- task_17_customer_age.png
- task_17_customer_gender.png

2.1.15 Risk Assessment Analysis

2.1.15.1 Insight:

- Risk varies across loan purpose
- Credit score strongly impacts risk

2.1.15.2 Visualization:



Note: charts reference:

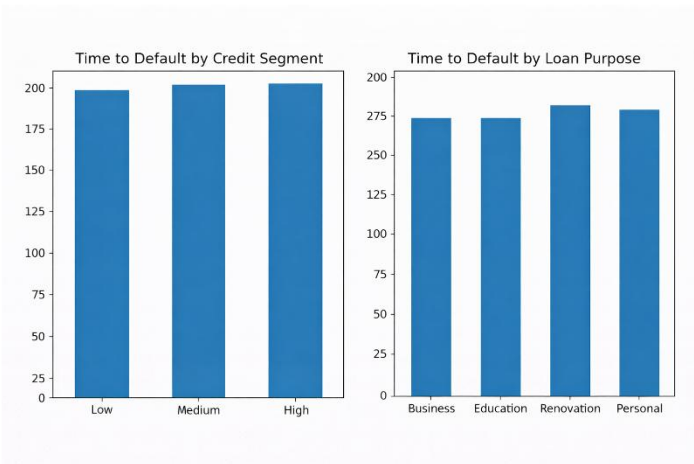
- task_17_customer_behavior.png
- task_17_customer_age.png
- task_17_customer_gender.png

2.1.16 Time to Default Assessment Analysis

2.1.15.1 Insight:

- Some loans default faster
- Faster default mean higher risk exposure

2.1.15.2 Visualization:



Note: charts reference:

- task_19_time_to_default_purpose.png
- task_19_time_to_default_credit.png

2.2 Visualization Summary

All the charts can be found under **reports/figures** in project structure

Purpose:

- Identify patterns
- Detect anomalies
- Support decision-making

Charts:

- Distribution Plots charts
- Trend Plots charts
- Time-series analysis charts
- Bar Charts

2.3 Recommendation

1. Reduce Loan Defaults.
2. Operation optimization.
3. Focus on improve Recovery.
4. Make Customer Strategies.
5. Data Improvement: Add BRANCH_ID linkage to loan and other data set.

3. Project Architecture

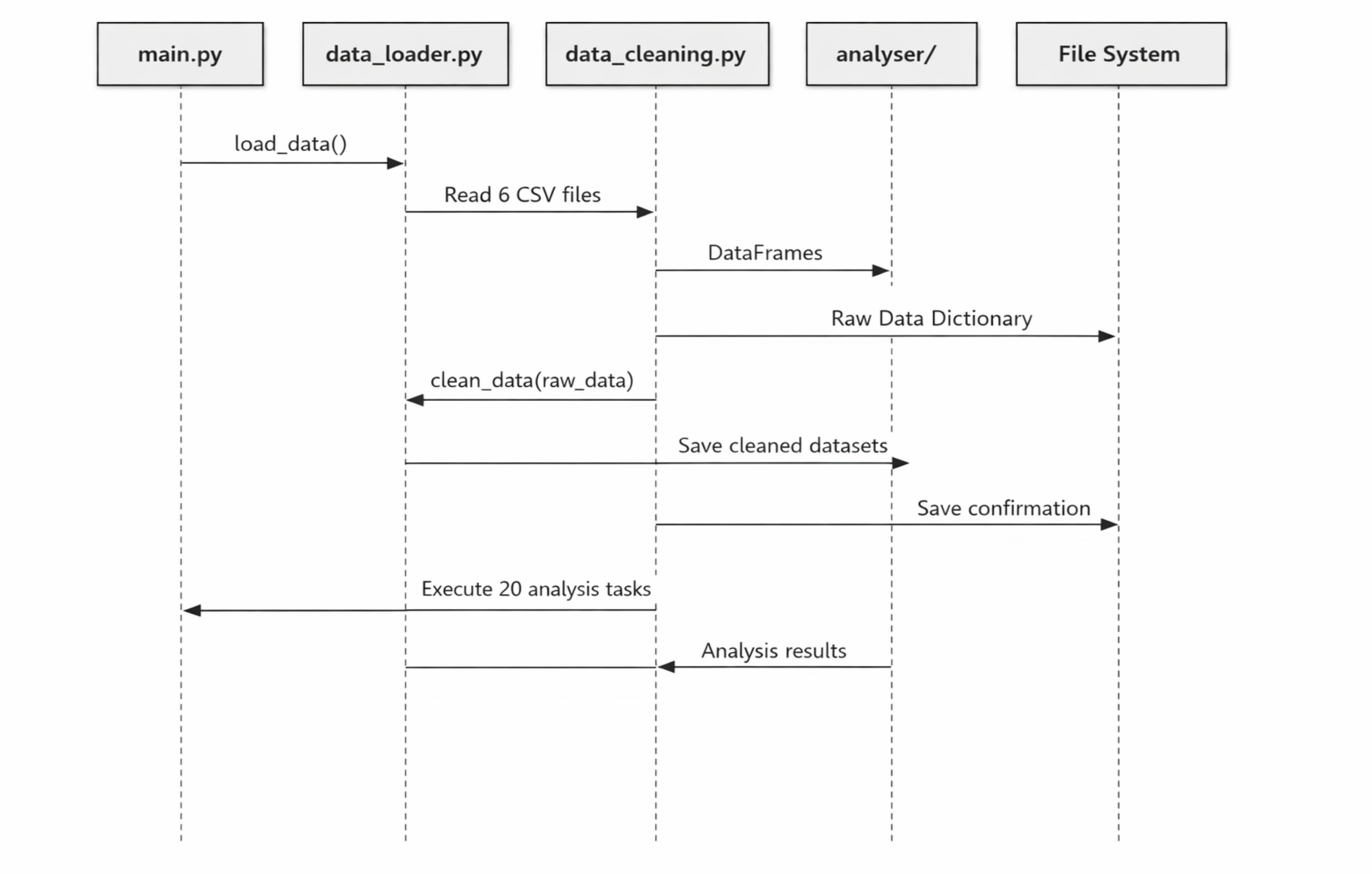
Below is a comprehensive architecture documentation for the Hero FinCorp data analysis project. Here's the complete architectural overview:

3.1 High-Level Architecture

The project follows a **layered architecture pattern** with 5 main layers:

1. **Data Layer** - Raw, cleaned, and processed data storage
2. **Processing Layer** - Data loading, cleaning, feature engineering, and merging
3. **Analysis Layer** - 20 analysis modules covering different aspects
4. **Visualization Layer** - Plot generation and chart creation
5. **Reporting Layer** – (Not implemented currently it is for later enhancement)Word, Excel, and visual report generation

3.2 Main Pipeline Sequence Diagram



3.3 Directory Structure

```
pythonassignment/
├── 🗎 main.py                    # Main execution pipeline
├── 🗎 config.py                 # Configuration settings
├── 🗀 data/                     # Data storage
│   ├── 🗀 raw/                  # 6 CSV files (customers, loans, applications,
transactions, defaults, branches)
│   ├── 🗀 cleaned/              # Processed clean data
│   └── 🗀 processed/            # Feature engineered data
├── 🗀 src/                      # Source code
│   └── 🗀 pgds/assignment/
│       ├── 🗀 dataprocessor/    # Data processing modules
│       ├── 🗀 analyser/         # 20 analysis modules
│       ├── 🗀 visualizer/       # Visualization modules
│       └── 🗀 reporting/        # Reporting modules
└── 🗀 reports/                  # Output reports and visualizations
```

3.4 Application Architecture Pattern

Layered Architecture with clear separation of concerns:

- **Presentation Layer**: main.py orchestration
- **Business Logic Layer**: analyser/ modules
- **Data Access Layer**: dataprocessor/ modules
- **Data Layer**: CSV files and processed data